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DB Wrong Ones

Q1: BCNF is not used for cases where a relation has

1. Two (or more) candidate keys
2. Two candidate keys and composite
3. The candidate key overlap
4. Two mutually exclusive foreign keys

Answer: 4. Two mutually exclusive foreign keys

Explanation:

A relation is in Boyce-Codd normal form if all attributes which are determinants are also candidate keys.
Transformation into Boyce-Codd normal form deals with the problem of overlapping keys.

Q4: Consider the schemaR = {S, T, U, V} and the dependencies
S → T, T → U, U → V and V → S If R = (R1 and R2)
be a decomposition such that R1 ∩ R2 = ϕ then the decomposition is

Options:
1. not in 2NF
2. in 2NF but not in 3NF
3. in 3NF but not in 2NF
4. in both 2NF and 3NF

Answer: 1. not in 2NF
Explanation:
Decomposition of a relation R into R1 and R2 is said to be lossless only if R1 ∩ R2 ≠ ϕ and the common attribute of R1 and R2 should be the primary key in any of the two decomposed relations(i.e.R1 and R2).
A relation which is in 2NF is always lossless.
In question, it is given that R1 ∩ R2 = ϕ which is violating lossless decomposition condition.
Hence the given relation R is not in 2NF.

Consider the table below:

employee

fsname	lname	ssl	bdate	address	sex	salary	s_ssl	dno
Rita	Chatterjee	123	1965-01-09	731 Fondren	F	30000	333	5
Rajeev	Sanyal	333	1955-12-08	638 Voss	M	40000	888	5
Sanjay	Agrawal	999	1978-01-19	3321 Castle	M	25000	987	4
Jennifer	Agacy	987	1980-12-08	291 Voss	F	45000	888	4
Narayan	Chaudhuri	666	1980-10-04	561 Castle	M	47000	333	5
Aishwarya	Kapoor	453	1972-09-08	291 Oak	F	25000	333	5
Mahesh	Srivastava	988	1982-11-07	567 Oak	M	35000	987	4
James	Stanley	888	1962-10-01	467 Oak	M	56000	⊥	1

The data types of the columns of the table `employee` are given below:

- `fsname, lname, bdate, address, sex: VARCHAR`
- `ssl, salary, s_ssl, dno: INT`

How many rows of `employee` table will be selected for the query given below?

```
SELECT E.salary
FROM employee AS  E, employee AS S
WHERE E.s_ssl = S.ssl;
```

a) 4

b) 2

c) 7

d) 8

Consider the table below:

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The data types of the columns of the table `employee` are given below:

- `fsname, lname, bdate, address, sex: VARCHAR`
- `ssl, salary, s_ssl, dno: INT`

What will be the output of the given query?

```
SELECT lname
FROM employee
WHERE fsname = 'Jennifer' OR  s_ssl = 987 AND address like '7%';
```

a) Agacy

b) Chatterjee

c) Srivastava

d) Agrawal

9) Consider the following grant statement

```
GRANT
   CREATE INDEX
TO
   Sahana
WITH ADMIN OPTION;
```

Identify the correct statement.

a) Sahana can only create new indexes but not grant this privilege to any user

b) Sahana can not only create index but also create tables, update and select them

c) Sahana can grant create index privilege to other users

d) WITH ADMIN is used to provide object level privileges